

Prob 1 - FDM for waveeq.

$$u_{tt} = 9 u_{xx} \quad -2 \leq x \leq 2, \quad t > 0$$

$$u(-2, t) = u(2, t) \quad t \geq 0$$

$$u(x, 0) = F(x), \quad u_x(x, 0) = g(x) \quad -2 \leq x \leq 2$$

a) $F(x) = \sin\left(\frac{\pi x}{2}\right) \quad \wedge \quad g(x) = 0$

$$u(x, t) = F(x)G(t)$$

$$F(x)G''(t) = 9F''(x)G(t)$$

$$\frac{F''(x)}{F(x)} = \frac{G''(t)}{9G(t)} = k$$

$$F''(x) - kF(x) = 0 \quad , \quad k = -p^2$$

$$F''(x) + p^2 F(x) = 0 \Rightarrow F(x) = A \cos(px) + B \sin(px)$$

$$F(-2) = A \cos(2p) - B \sin(2p) = 0 \quad , \quad p = \frac{n\pi}{4}$$

$$F(2) = A \cos(2p) + B \sin(2p) = 0$$

$$F(2) + F(-2) = A \cos(2p) = 0 \Rightarrow A = 0 \quad \wedge \quad p = \frac{2n\pi}{4} \quad , \quad n = 1, \dots$$

$$F(2) - F(-2) = 2B \sin(2p) = 0 \Rightarrow B = 0 \quad \wedge \quad p = \frac{(2n+1)\pi}{4} \quad n = 0, \dots$$

$$F_n(x) = B_n \sin\left(\frac{n\pi x}{2}\right) \quad (F_n(x) = A \cos\left(\frac{(2n+1)\pi}{4} x\right))$$

$$G_n(t) = C_n \cos\left(\frac{3n\pi}{2} t\right) + D_n \sin\left(\frac{3n\pi}{2} t\right)$$

$$u_n(x, t) = F_n(x)G_n(t) \Rightarrow u(x, t) = \sum F_n G_n$$

Case 1: $u(x, 0) = \sum \sin\left(\frac{n\pi x}{2}\right) = F(x) = \sin\left(\frac{\pi x}{2}\right) \Rightarrow B_1 = 1$

$$u_t(x, t) = \frac{3n\pi}{2} \sin\left(\frac{n\pi x}{2}\right) (D_n \cos\left(\frac{3n\pi}{2} t\right) - C_n \sin\left(\frac{3n\pi}{2} t\right))$$

$$u_t(x, 0) = \frac{3n\pi}{2} B_n \sin\left(\frac{n\pi x}{2}\right) = 0 \Rightarrow B_n = 0 \quad \forall n$$

$$u(x, t) = \cos\left(\frac{3\pi}{2} t\right) \sin\left(\frac{\pi x}{2}\right) \quad , \quad n = 1 \Rightarrow u(x, t) = \cos\left(\frac{3\pi}{2} t\right) \sin\left(\frac{\pi x}{2}\right)$$

Case 2: $u(x, 0) = \sum \sin\left(\frac{n\pi x}{2}\right) = F(x) = 0 \Rightarrow B_n = 0 \quad \forall n$

$$u_t(x, t) = \frac{3n\pi}{2} \sin\left(\frac{n\pi x}{2}\right) (D_n \cos\left(\frac{3n\pi}{2} t\right) - C_n \sin\left(\frac{3n\pi}{2} t\right))$$

$$u_t(x, 0) = \frac{3n\pi}{2} B_n \sin\left(\frac{n\pi x}{2}\right) = \pi \sin\left(\frac{\pi x}{2}\right) \Rightarrow B_1 = \frac{2}{3}$$

$$u(x, t) = \frac{2}{3} \sin\left(\frac{3\pi}{2} t\right) \sin\left(\frac{\pi x}{2}\right)$$

b)

$$u(x+h, t) = u(x, t) + u'(x, t)h + \frac{u''(x, t)h^2}{2}$$

$$u(x-h, t) = u(x, t) - u'(x, t)h + \frac{u''(x, t)h^2}{2}$$

$$u(x+h, t) + u(x-h, t) = u(x, t) + u'(x, t)h + \frac{u''(x, t)h^2}{2} + (u(x, t) - u'(x, t)h + \frac{u''(x, t)h^2}{2})$$

$$u''(x, t) = \frac{u(x+h, t) + u(x-h, t) - 2u(x, t)}{h^2} = \frac{U_{n+1}^n + U_{n-1}^n + U_n^n}{h^2}$$

$$u(x, t+k) = u(x, t) + u'(x, t)k + \frac{u''(x, t)k^2}{2}$$

$$u(x, t-k) = u(x, t) - u'(x, t)k + \frac{u''(x, t)k^2}{2}$$

$$u(x, t+k) + u(x, t-k) = u(x, t) + u'(x, t)k + \frac{u''(x, t)k^2}{2} + (u(x, t) - u'(x, t)k + \frac{u''(x, t)k^2}{2})$$

$$u''(x, t) = \frac{u(x, t+k) + u(x, t-k) - 2u(x, t)}{k^2} = \frac{U_{n+1}^{n+1} + U_{n-1}^{n+1} + 2U_n^{n+1}}{k^2}$$

$$\frac{U_{n+1}^{n+1} + U_{n-1}^{n+1} + 2U_n^{n+1}}{k^2} = 9 \frac{U_{n+1}^n + U_{n-1}^n + 2U_n^n}{h^2}$$

$$\frac{U_{n+1}^{n+1}}{k^2} + \frac{U_{n-1}^{n+1}}{k^2} + \frac{2U_n^{n+1}}{k^2} = \frac{9U_{n+1}^n}{h^2} + \frac{9U_{n-1}^n}{h^2} + \frac{18U_n^n}{h^2}$$

$$U_{n+1}^{n+1} + U_{n-1}^{n+1} + 2U_n^{n+1} = k^2 \frac{9U_{n+1}^n}{h^2} + k^2 \frac{9U_{n-1}^n}{h^2} + k^2 \frac{18U_n^n}{h^2}$$

$$U_{n+1}^{n+1} = \frac{9k^2}{h^2} (U_{n+1}^n + U_{n-1}^n) + U_n^n \left(\frac{18k^2}{h^2} - 2 \right) - U_{n-1}^{n+1}$$

The given Dirichlet conditions in (a) gives us $u_r(x, 0) = 0$

$$U_1^{1+1} = \frac{9k^2}{h^2} (U_2^0 + U_0^0) + U_1^0 \left(\frac{18k^2}{h^2} - 2 \right) - U_1^{0+1}$$

$$U_{n+1}^{n+1} = \frac{9k^2}{h^2} (U_{n+1}^0 + U_{n-1}^0) + U_n^0 \left(\frac{18k^2}{h^2} - 2 \right) - U_{n-1}^{n+1}$$

$$u(x, t+k) - u(x, t-k) = \cancel{u(x, t)} + u'(x, t)k + \frac{\cancel{u''(x, t)k^2}}{2} - (\cancel{u(x, t)} - u'(x, t)k + \frac{\cancel{u''(x, t)k^2}}{2})$$

$$u'(x, t) = \frac{u(x, t+k) - u(x, t-k)}{2k} = \frac{U_{n+1}^{n+1} - U_{n-1}^{n+1}}{2k} \Rightarrow u_r(x, 0) = 0 \Rightarrow \frac{U_m^1 - U_m^{-1}}{2k} = 0 \Rightarrow U_m^1 = U_m^{-1}$$

$$n=0: U_m^1 = \frac{9k^2}{h^2} (U_{m+1}^0 + U_{m-1}^0) + U_m^0 \left(\frac{18k^2}{h^2} - 2 \right) - U_m^{-1}$$

$$U_m^1 = \frac{9k^2}{h^2} (U_{m+1}^0 + U_{m-1}^0) + U_m^0 \left(\frac{18k^2}{h^2} - 2 \right) - U_m^1$$

$$2U_m^1 = \frac{9k^2}{h^2} (U_{m+1}^0 + U_{m-1}^0) + U_m^0 \left(\frac{18k^2}{h^2} - 2 \right)$$

$$m=1, h=1, k=\frac{1}{3} U_1^1 = \frac{1}{2} \left(\frac{9(1/3)^2}{1} (U_2^0 + U_0^0) + U_1^0 \left(\frac{18(1/3)^2}{1} - 2 \right) \right)$$

$$= \frac{1}{2} ((u(2, 0) + u(0, 0)) + u(1, 0)(2-2))$$

$$= \frac{1}{2} (f(2) + f(0)) = 0$$

$$m=2, \quad u - u_2^1 = \frac{1}{2} \left(\frac{9(1/3)^2}{1} (U_3^0 + U_1^0) + U_2^0 \left(\frac{18(1/3)^2}{1} - 2 \right) \right) = 0$$

$$m=3, \quad U_3^1 = 0$$

Prob 2

$$u_{tt} = 4 u_{xx} \quad , \quad u(x,0) = \overset{=f(x)}{\sin(x)} \quad , \quad u_t(x,0) = \overset{=g(x)}{e^{-\frac{x}{4}}}$$

$$u(x,t) = \frac{1}{2} (f(x+ct) + f(x-ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\tilde{x}) d\tilde{x}$$

$$= \frac{1}{2} (\sin(x+2t) + \sin(x-2t)) + \frac{1}{4} \int_{x-2t}^{x+2t} e^{-\frac{\tilde{x}}{4}} d\tilde{x}$$

$$= \sin(x) \cos(2t) + \frac{1}{4} [-4e^{-\frac{\tilde{x}}{4}}]_{x-2t}^{x+2t}$$

$$= \sin(x) \cos(2t) + (-e^{-\frac{1}{4}(x+2t)} - (-e^{-\frac{1}{4}(x-2t)}))$$

$$= \sin(x) \cos(2t) + (e^{-\frac{1}{4}(x-2t)} - e^{-\frac{1}{4}(x+2t)})$$

Prob 3 - 1D Heat eq.

$$u_t = \frac{1}{4} u_{xx} \quad , \quad u(0,t) = u(\pi/2,t) = 0 \quad , \quad u(x,0) = 2x(x-\pi/2)$$

$$u(x,t) = F(x)G(t)$$

$$F(x)G'(t) = \frac{1}{4} F''(x)G(t)$$

$$\frac{F''(x)}{F(x)} = \frac{G'(t)}{\frac{1}{4}G(t)} = k$$

$$F''(x) - kF(x) = 0$$

$$k=0 \Rightarrow F(x) = Ax + B \Rightarrow \text{initialbet gir da } A=B=0 \Rightarrow F(x)=0$$

$$k>0 \Rightarrow F(x) = Ae^{\sqrt{k}x} + Be^{-\sqrt{k}x} \Rightarrow \text{initialbet gir da } A=B=0 \Rightarrow F(x)=0$$

$$\text{Altså må } k<0 \Rightarrow \text{skriver } k=-p^2 \text{ for å sikre neg.}$$

$$F(x) = A \cos(px) + B \sin(px) \quad , \quad p = \frac{n\pi}{L}$$

$$F(0) = A \cos(0) = 0 \Rightarrow A=0$$

$$F(\pi/2) = B \sin(p\pi/2) = 0 \quad , \quad B \neq 0 \Rightarrow p = \frac{n\pi}{L} = 2n$$

$$F_n(x) = B_n \sin(\frac{2n\pi}{L}x) = B_n \sin(4nx) = B_n \sin(2nx)$$

$$G_n(t) = G_0 e^{-\frac{(p\pi)^2 t}{4}} = G_0 e^{-\frac{(2n)^2 \pi^2 t}{4}} = G_0 e^{-n^2 \pi^2 t}$$

$$u_n(x,t) = \sum_{n=1}^{\infty} B_n \sin(2nx) e^{-n^2 \pi^2 t}$$

$$B_n = \frac{2}{L} \int_0^L F(x) \sin(\frac{n\pi x}{L}) dx \quad , \quad L = \pi/2$$

$$= \frac{2}{\pi/2} \int_0^{\pi/2} (2x^2 - \pi x) \sin(2nx) dx$$

$$= \frac{4}{\pi} \int_0^{\pi/2} x^2 \sin(2nx) dx - 2 \int_0^{\pi/2} x \sin(2nx) dx$$

$$\begin{aligned}
\frac{1}{2} \int_0^{\frac{\pi}{2}} u v' &= uv - \int u' v dx \\
\frac{1}{2} \int_0^{\frac{\pi}{2}} x^2 \sin(2nx) dx &= \left[-\frac{x^2 \cos(2nx)}{2n} + \frac{1}{n} \int x \cos(2nx) dx \right]_0^{\frac{\pi}{2}} \\
&= \left[-\frac{x^2 \cos(2nx)}{2n} + \frac{1}{n} \left(\frac{x \sin(2nx)}{2n} - \frac{1}{2n} \int \sin(2nx) dx \right) \right]_0^{\frac{\pi}{2}} \\
&= \left[-\frac{x^2 \cos(2nx)}{2n} + \frac{1}{n} \left(\frac{x \sin(2nx)}{2n} + \frac{\cos(2nx)}{4n^2} \right) \right]_0^{\frac{\pi}{2}} \\
&= \left[-\frac{x^2 \cos(2nx)}{2n} + \frac{x \sin(2nx)}{2n^2} + \frac{\cos(2nx)}{4n^2} \right]_0^{\frac{\pi}{2}} \Rightarrow \frac{1}{4n^2} \\
&= -\frac{\pi^2 \cos(n\pi)}{8n} + \frac{\pi \sin(n\pi)}{4n^2} + \frac{\cos(n\pi)}{4n^2} - \frac{1}{4n^2} \\
&= -\frac{\pi^2 (-1)^n}{8n} + \frac{(-1)^n}{4n^2} - \frac{1}{4n^2} \cdot \frac{4}{\pi} \\
&= -\frac{\pi (-1)^n}{2n} + \frac{(-1)^n - 1}{4n^2}
\end{aligned}$$

$$\begin{aligned}
\frac{1}{2} \int_0^{\frac{\pi}{2}} x \sin(2nx) dx &= \left[-\frac{x \cos(2nx)}{2n} + \frac{1}{2n} \int \cos(2nx) dx \right]_0^{\frac{\pi}{2}} \\
&= \left[-\frac{x \cos(2nx)}{2n} + \frac{\sin(2nx)}{4n^2} \right]_0^{\frac{\pi}{2}} \Rightarrow 0 \\
&= -\frac{\pi \cos(n\pi)}{2n} \cdot 2 \\
&= -\frac{\pi \cos(n\pi)}{2n}
\end{aligned}$$

$$B_n = -\frac{\pi (-1)^n}{2n} + \frac{(-1)^n - 1}{4n^2} - \left(-\frac{\pi \cos(n\pi)}{2n} \right) = \frac{(-1)^n - 1}{4n^2}$$

Prob 4 - d'Alembert's

$$u_{tt} = u_{xx}, \quad u(x,0) = \cos(x), \quad u_t(x,0) = x e^{-x^2}$$

$$\begin{aligned}
u(x,t) &= \frac{1}{2} (f(x+ct) + f(x-ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\tilde{x}) d\tilde{x} \\
&= \frac{1}{2} (\cos(x+t) + \cos(x-t) + \int_{x-t}^{x+t} x e^{-x^2} dx)
\end{aligned}$$

$$u = -x^2 \quad \frac{du}{dx} = -2x \quad dx = -\frac{du}{2x}$$

$$= \cos(x) \cos(t) + \frac{1}{2} \int_{x-t}^{x+t} x e^{-x^2} \left(-\frac{du}{2x} \right)$$

$$= \cos(x) \cos(t) - \frac{1}{4} \int_{x-t}^{x+t} e^{-u} du$$

$$= \cos(x) \cos(t) - \frac{1}{4} [e^{-u}]_{x-t}^{x+t}$$

$$= \cos(x) \cos(t) - \frac{1}{4} (e^{-(x+t)^2} - e^{-(x-t)^2})$$